

	M.A.M. SCHOOL OF ENGINEERING	
	(An Autonomous Institution) Accredited by NAAC & Approved by AICTE, New Delhi Affiliated to Anna University, Chennai	
	Siruganur, Trichy -621 105.	www.mamse.in

Department of Electrical and Electronics Engineering
Academic Year 2025-2026(Even Semester)

Unit - I

Sub.Code/Sub.Name : EE3023/MEMS and NEMS
Year/Sem. : III/VI

Duration : 3 Hrs
Max. Marks : 100

	Part A (10 X 2 = 20 Marks) Answer All the Questions	M.A	CO	BTL
1.	What is MEMS?	2	CO1	BTL1
2.	List out the materials are used in MEMS	2	CO1	BTL1
3.	What is NEMS?	2	CO1	BTL1
4.	Recall the major energy domains.	2	CO1	BTL1
5.	Define 1/f noise.	2	CO1	BTL1
6.	List the advantages of NEMS	2	CO1	BTL1
7.	Define the miniaturization.	2	CO1	BTL1
8.	Write the components of MEMS	2	CO1	BTL1
9.	State the range of motion.	2	CO1	BTL1
10.	Define sensor reliability.	2	CO1	BTL1

		Part B (5 X 16 =80)	M.A	CO	BTL
11.	a)	Elucidate the history of MEMS development.	8	CO1	BTL2
	b)	Discuss the Future trends of MEMS.	8	CO1	BTL2
12.	a)	Briefly write the history of NEMS development.	8	CO1	BTL2
	b)	Explain the Sensor considerations of MEMS	8	CO1	BTL2
13.	a)	Summarize the energy domain transducer.	8	CO1	BTL2
	b)	Explain the application of MEMS development and explain	8	CO1	BTL2
14.	a)	Describe the NEMS development.	8	CO1	BTL2
	b)	Discuss the fabricated process for NEMS devices.	8	CO1	BTL2
15.	a)	Explain the Law of scaling materials in MEMS	8	CO1	BTL2
	b)	Write the application of NEMS development.	8	CO1	BTL1

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Unit - II

Sub.Code/Sub.Name : EE3023/MEMS and NEMS
Year/Sem. : III /VI

Duration : 3 Hrs
Max. Marks : 100

	Part A (10 X 2 = 20 Marks) Answer All the Questions	M.A	CO	BTL
1.	Difference between dry and wet etching.	2	CO2	BTL1
2.	State LIGA process.	2	CO2	BTL1
3.	Define surface micromachining.	2	CO2	BTL1
4.	What is electrostatic sensor?	2	CO2	BTL1
5.	Explain pull-in effect of an electrostatic actuator?	2	CO2	BTL1
6.	List of the methods of making alignment	2	CO2	BTL1
7.	Recall the parallel plate capacitive accelerometer.	2	CO2	BTL1
8.	List some Thermal properties of common materials	2	CO2	BTL1
9.	What is miniaturization?	2	CO2	BTL1
10.	Write the thermal properties of common materials?	2	CO2	BTL1

		Part B (5 X 16 =80)	M.A	CO	BTL
11.	a)	Briefly explain the photolithography.	8	CO2	BTL2
	b)	Explain the photolithography fabrication process and list the application also.	8	CO2	BTL2
12.	a)	Summarize the bulk micromachining fabrication process	8	CO2	BTL2
	b)	Discuss the bulk Micro machined SMA Actuator	8	CO2	BTL2
13.	a)	Explain the surface micromachining fabrication process.	8	CO2	BTL2
	b)	Draw and explain the surfaced micro machining.	8	CO2	BTL2
14.	a)	Explain LIGA fabrication process and list it's advantages, disadvantages and application.	16	CO2	BTL2
15.	a)	List out the micro fabrication process in step by step.	8	CO2	BTL1
	b)	What are the Etchants for common micro fabrication materials	8	CO2	BTL1

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Unit - III

Sub.Code/Sub.Name : EE3023/MEMS and NEMS
Year/Sem. : III/VI

Duration : 3 Hrs
Max. Marks : 100

	Part A (5 X 2 = 10 Marks) Answer All the Questions	M.A	CO	BTL
1.	Define PVDF	2	CO3	BTL1
2.	List the values are piezoresistive components for single crystal silicon under certain doping.	2	CO3	BTL1
3.	Write the application of piezoresistive sensor	2	CO3	BTL1
4.	Why silicon and germanium cannot be used as piezoelectric sensors?	2	CO3	BTL1
5.	What is meant by centre symmetric crystal?	2	CO3	BTL1
6.	What factors influence the performance of capacitive sensors?	2	CO3	BTL1
7.	Name some bio medical applications of piezo electric actuators?	2	CO3	BTL1
8.	List some challenges in the fabrication of capacitive sensors.	2	CO3	BTL1
9.	What are the limitations of capacitive sensors?	2	CO3	BTL1
10.	What are some techniques used to improve the sensitivity of piezo resistive sensors?	2	CO3	BTL1

		Part B (2 X 16 =32)	M.A	CO	BTL
11.	a)	Discuss the applications of piezoresistive sensors in MEMS.	8	CO3	BTL2
	b)	Elucidate piezoresistive sensor used in MEMS	8	CO3	BTL2
12.	a)	Discuss about the piezoelectric actuators and its application.	8	CO3	BTL2
	b)	Describe the concept of Piezoelectric effects	8	CO3	BTL2
13.	a)	Explain the working principle of capacitive sensors	8	CO3	BTL2
14.	a)	Discuss the advantages and limitations of microsensors in industrial applications.	8	CO3	BTL2
15.	a)	Discuss the challenges and opportunities in developing microactuators for biomedical applications.	8	CO3	BTL2

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**Department of Electrical and Electronics Engineering
Academic Year 2025-2026(Even Semester)**

Unit - IV

Sub.Code/Sub.Name : EE3023/MEMS and NEMS
Year/Sem. : III / VI

Date : 13.05.2025
Duration : 3 Hrs
Max. Marks : 100

Part A (10 X 2 = 20 Marks) Answer All the Questions		M.A	CO	BTL
1.	Define NEMS Technology.	2	CO4	BTL1
2.	What are the common NANO Fabrication techniques?	2	CO4	BTL1
3.	List the types of sensors can be made using NEMS.	2	CO4	BTL1
4.	How do NEMS device benefit from atomic scale precision?	2	CO4	BTL1
5.	What role does NANO technology play in advancing NEMS technology?	2	CO4	BTL1
6.	How do NEMS devices contribute to the field of robotics?	2	CO4	BTL1
7.	Give some environmental sensing applications of NEMS technology.	2	CO4	BTL1
8.	How do NEMS devices address the need for miniaturization in electronics?	2	CO4	BTL1
9.	What is the significance of nano- scale features in NEMS devices?	2	CO4	BTL1
10.	What are some emerging trends in NEMS technology research?	2	CO4	BTL1

Part B (5 X 16 =80)		M.A	CO	BTL
11.	a) Explain the following (i) Atomic Force Microscopy. (ii) Scanning Tunneling microscopy. (iii) Atomic Layer Deposition.	16	CO4	BTL2
12.	a) Describe the Electrostatic and Electro Thermal functions for Nano actuators.	13	CO4	BTL2
13.	a) Discuss the NEMS Based Sensing techniques for piezo-resistive, Capacitive, and Thermal.	13	CO4	BTL2
14.	a) Generalize the different types of Nano fabrication methods.	13	CO4	BTL2
15.	a) Explain the Different types of Nano partical in Nems fabrication.	13	CO4	BTL2

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Unit - V

Sub.Code/Sub.Name : EE3023/MEMS and NEMS
Year/Sem. : III / VI

Duration : 3 Hrs
Max. Marks : 100

	Part A (10 X 2 = 20 Marks) Answer All the Questions	M.A	CO	BTL
1.	Give an example of a Bio MEMS application.	2	CO5	BTL1
2.	What are Optical NEMS?	2	CO5	BTL1
3.	what is a Micro Motor?	2	CO5	BTL1
4.	How are smart sensors different from traditional sensors?	2	CO5	BTL1
5.	Give some recent trends in MEMS and NEMS.	2	CO5	BTL1
6.	List the advantages do Micro Motors offer in robotics.	2	CO5	BTL1
7.	How do Optical NEMS contribute to telecommunications?	2	CO5	BTL1
8.	What role do smart sensors play in the development to smart cities?	2	CO5	BTL1
9.	What are some emerging application of NEMS in quantum technology?	2	CO5	BTL1
10.	How do MEMS sensors contribute to the Internet of Things(IoT)?	2	CO5	BTL1

		Part B (5 X 16 =80)	M.A	CO	BTL
11.	a)	Describe the working principle of a Bio mems devices.	13	CO5	BTL2
12.	a)	Discuss the future prospects of MEMS and NEMS in industrial and consumer applications.	13	CO5	BTL2
13.	a)	Explain the working of Micro Motors, and also list its advantage, disadvantage and limitation	13	CO5	BTL2
14.	a)	Describe The design challenges and opportunities in developing Optical NEMS devices	13	CO5	BTL2
15.	a)	Discuss the potential applications of Smart Sensors in IOT and Other emerging technologies.	13	CO5	BTL1